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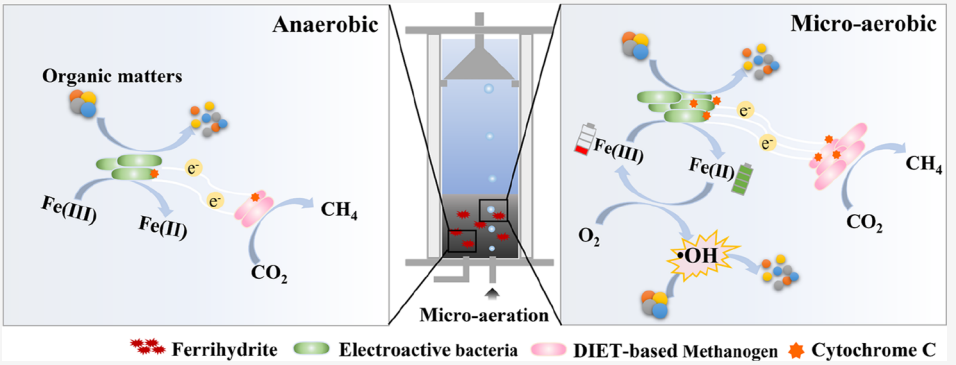
Intermittent Microaeration Enhanced Anaerobic Digestion: The Key Role of Fe(III)/Fe(II) Cycle and Reactive Oxygen Species

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ABSTRACT: Microaeration has been reported to improve anaerobic digestion, which is generally attributed to increased microbial diversity, but a thorough explanation is lacking. In this study, it was found that intermittent microaeration (IMA) supplied to an anaerobic digester could oxidize Fe(II) produced by dissimilatory iron reduction (DIR) to form a Fe(III)/Fe(II) cycle and generate extracellular reactive oxygen species (ROS) to improve anaerobic treatment of phenol-containing wastewater. The results showed that compared to the control group without IMA, the removal rates of COD and phenol increased by 29.54 and 49.68 percentage points, respectively, and daily average methane production increased by 85.44%. The Fe(III)/Fe(II) cycle slowed down the loss of iron (13.97%) released from sludge due to the lower solubility of Fe(III) and facilitated •OH generation (1.22 ± 0.04 μM) via Fenton-like reactions. The DIR and the generation of •OH accelerated phenol degradation. Metagenomic analysis revealed that the abundance of methanogens and antioxidant enzymes-encoding genes in response to oxidative stress significantly increased in the IMA group compared to the control, enabling methanogenesis to proceed smoothly under microaeration. This study investigated the extracellular ROS generation induced by microaeration during anaerobic digestion and their roles in promoting anaerobic performance, thereby providing a new perspective for optimizing anaerobic systems with microaeration.

KEYWORDS: *dissimilatory iron reduction, Fe(III)/Fe(II) cycle, reactive oxygen species, antioxidant defense, direct interspecies electron transfer*

1. INTRODUCTION

Anaerobic digestion is considered one of the most widely used technologies to treat high-concentration organic wastewater due to its effectiveness in recovering biomass energy (methane) from organic matter.[1,2] However, anaerobic digestion is challenged by low operational stability and slow hydrolytic acidification of complex substrates. Additionally, numerous toxic organic pollutants, such as phenol in industrial organic wastewater, impede anaerobic digestion and methane production.[3,4] Phenol is widely present in industrial wastewater from coal chemical processing, pharmaceutical manufacturing, and other industries, and has been listed as one of the priority control pollutants by the United States Environmental Protection Agency due to its high toxicity.[5] It has been reported that phenol over 2000 mg/L inhibits biological processes including anaerobic digestion due to the toxic accumulation of phenol.[6,7] Consequently, accelerating the decomposition of phenol is essential for improving the anaerobic treatment of phenol-containing wastewater.

The addition of iron (hydro)oxides has been extensively reported to promote the degradation of complex organic matter through microbially dissimilatory iron reduction (DIR).[8,9] It also can facilitate the syntrophic metabolism between dissimilatory iron-reducing bacteria and methanogens capable of direct interspecies electron transfer (DIET) to increase methane production.[3,10] DIR, also known as iron

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respiration, refers to microorganisms using organic matter as electron donors and extracellular insoluble iron (hydro)oxides as terminal electron acceptors to reduce Fe(III) and produce dissolved Fe(II), accompanied by energy output for microbial growth. However, Fe(III) has a low electron equivalent, theoretically necessitating a constant and large supplementation of iron (hydro)oxides to sustain a continuous DIR, which is impractical in practice. If the Fe(II) produced by DIR in the anaerobic process can be oxidized to Fe(III) to drive an Fe(III)/Fe(II) cycle, then this iron-based method relying on DIR is expected to be applied in practical engineering.

Many oxidants, such as O_2 and NO_2^-/NO_3^-, can oxidize Fe(II), resulting in the widespread occurrence of the Fe(III)/Fe(II) cycle in natural water systems.[11,12] Notably, O_2, as the most ubiquitous oxidant, can microbially oxidize Fe(II) by neutrophilic Fe(II)-oxidizing bacteria, which grow autotrophically by utilizing Fe(II) as an electron donor and O_2 as the electron acceptor (reaction 1):[12]

$$4Fe^{2+} + 10H_2O + O_2 \rightarrow 4Fe(OH)_3 + 8H^+ \quad (1)$$

Furthermore, at circumneutral pH, Fe(II) may be chemically oxidized to Fe(III) ($E^{\theta}(O_2/H_2O)$ = 1.23 V, and $E^{\theta}(Fe^{3+}/Fe^{2+})$ = 0.77 V) by O_2 through the following reactions 2−5. These processes are accompanied by the generation of reactive oxygen species (ROS, i.e., $^{\bullet}O_2^-$, H_2O_2, and $^{\bullet}OH$).[12]

$$Fe^{2+} + O_2 \rightarrow Fe^{3+} + {}^{\bullet}O_2^- \quad (2)$$

$$Fe^{2+} + {}^{\bullet}O_2^- + 2H^+ \rightarrow Fe^{3+} + H_2O_2 \quad (3)$$

$$Fe^{2+} + H^+ + H_2O_2 \rightarrow Fe^{3+} + {}^{\bullet}OH + H_2O \quad (4)$$

$$Fe^{2+} + {}^{\bullet}OH + H^+ \rightarrow Fe^{3+} + H_2O \quad (5)$$

On the other hand, from a traditional perspective, O_2 should be avoided as much as possible during anaerobic digestion to minimize its negative effects on obligate anaerobic microorganisms.[13] Nevertheless, recent studies have reported that microaeration can promote anaerobic digestion.[13,14] This might be attributed to the fact that microaeration enhances microbial diversity, which is beneficial for the hydrolysis of complex organic matter and the removal of hydrogen sulfide.[15−17] Also, the stirring effect of intermittent microaeration increases mass transfer.[18,19] However, it remains unclear whether the Fe(II) oxidation and ROS generation are involved in the promotion of anaerobic methanogenesis by intermittent microaeration.

Theoretically, intermittent microaeration is capable of reoxidizing Fe(II) to Fe(III) in anaerobic digestion systems, thereby driving the Fe(III)/Fe(II) cycle and creating a continuous DIR.[20,21] In this process, the generated ROS together with the Fe(II) oxidation are likely to attack the macromolecular organic matter,[22,23] and the sustained DIR caused by Fe(III) generation may accelerate the decomposition of the organic matter for methane production. In previous studies, intermittent aeration-driven Fe(III)/Fe(II) has been shown to promote anaerobic ammonium removal and persistent organic pollutant degradation.[20,24]

Several studies have reported ROS generation in anaerobic digesters.[25−27] However, most of these studies focus only on its adverse effects. That is to say, under unfavorable metabolic conditions, the accumulation of intracellular ROS may suppress metabolic activities, which in turn leads to the deterioration of the performance of anaerobic digesters.[28,29] For instance, Wei et al. documented the generation of intracellular ROS triggered by microplastics during the anaerobic digestion of waste-activated sludge, which significantly inhibited the digestion process.[26] This discrepancy hinders a more in-depth understanding of how the ROS induced by intermittent microaeration affects microbial processes. The impacts of ROS induced by intermittent microaeration on methanogenesis and anaerobic microbial metabolism remain ambiguous.

This study aimed to identify and validate the Fe(III)/Fe(II) cycle and ROS induced by intermittent microaeration during anaerobic methanogenesis. Furthermore, the influence of intermittent microaeration on the evolution of the microbial community was investigated.

2. MATERIALS AND METHODS

2.1. Inoculum Sludge and Ferrihydrite Preparation. The inoculum sludge was collected from the anaerobic tank of a municipal sludge treatment plant in Dalian, China, with a total solids (TSS) concentration of 39.75 g/L and volatile solids (VSS) of 36.65 g/L. Ferrihydrite was synthesized by neutralizing a 50 mM $FeCl_3$ solution with 0.5 M NaOH to a pH of about 7, as reported previously.[30] The morphology of the synthesized ferrihydrite is shown in Figure S1.

2.2. Operation of Reactors. Two parallel continuous-flow UASB reactors (φ70 mm × 300 mm, working volume: 1000 mL) were used for the anaerobic digestion. Each reactor was inoculated with 300 mL inoculum sludge and 700 mL of synthesized wastewater as the feeding substances. Besides, each reactor received the synthesized ferrihydrite with a final Fe(III) concentration of 15 mM. The initial composition of the synthesized wastewater (per liter) was as follows: 2.50 g glucose, 0.23 g NH_4Cl, 0.05 g KH_2PO_4, 2.50 g $NaHCO_3$, and 10 mL trace elements.[31] The hydraulic retention time was controlled at 30 h, and the reactor was operated at 33 ± 1 °C. One reactor received intermittent microaeration (referred to as the intermittent microaeration group, IMA group), with an aeration intensity of 14 mL $O_2/(L_{reactor}\cdot day)$, and the aeration was carried out for 30 min every 3 h. The other reactor served as a control (Control group) and was supplied with N_2 according to the same aeration schedule as that of the IMA group. The selection of ferrihydrite dosing and the intensity/frequency of microaeration are detailed in Text S1.

The continuous-flow experiments were carried out for 124 days. During the first 40 days, glucose was used as the sole carbon source, and the influent COD gradually increased from 2500 to 8000 mg/L. After that, glucose was gradually replaced by phenol as the substrate. The detailed variation of influent substrate during the operation is presented in Table S1.

2.3. ROS Measurement. The generation of $^{\bullet}OH$ was determined by an electron paramagnetic resonance (EPR) spectrometer and quantified using coumarin, as described in Text S2. The $^{\bullet}OH$ generation pathways were investigated by examining its intermediates (e.g., $^{\bullet}O_2^-$ and H_2O_2). To verify the pathway of $^{\bullet}OH$ generation, various ROS scavengers were added separately to the digester. Nitro blue tetrazolium (NBT) (5 mM) was employed to scavenge $^{\bullet}O_2^-$, and catalase (CAT) (1000 U/mL) was used to scavenge H_2O_2. Additionally, tert-butanol (TBA) and *n*-butanol (NBA) were utilized as $^{\bullet}OH$ quenchers.[23,32]

2.4. Characterization and Analysis. The morphology of the synthesized ferrihydrite was characterized by means of a scanning electron microscope and X-ray diffraction. The

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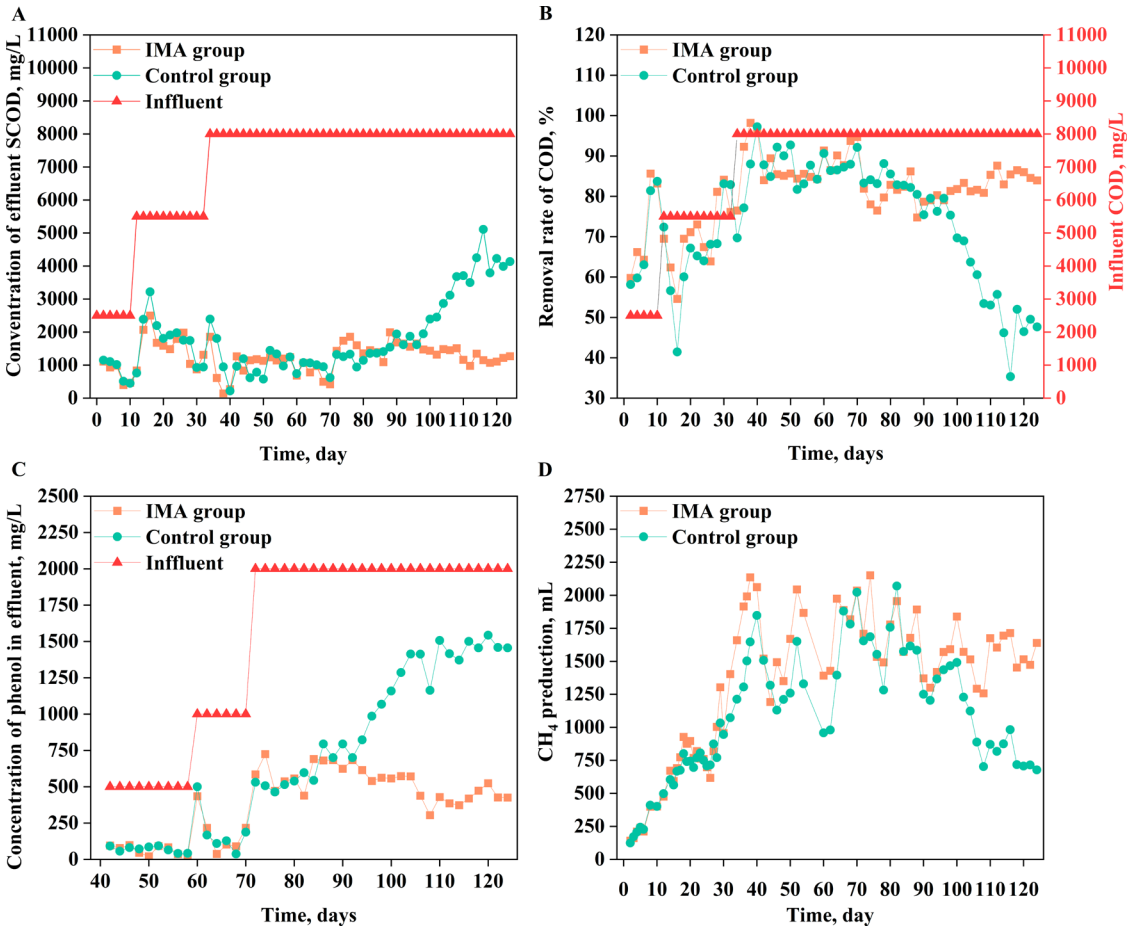


Figure 1. Performance of anaerobic digester. (A) Concentration of effluent COD. (B) Removal rate of COD. (C) Concentration of effluent phenol. (D) Daily methane production.

capacitance and resistance of the sludge were determined following the methods described in the reported study.[33]

The effluent from the anaerobic reactors was analyzed every 2 days. TSS, VSS, COD, the total content of ferrous and ferric iron, as well as the content and volume of biogas were analyzed as previously reported.[34,35] The phenol concentration was quantified by high-performance liquid chromatography (Shimadzu, LC-2050C, Japan) with a C18 column (5 μm, 250 μm × 4.6 mm). The mobile phase, with a flow rate of 1 mL/min, was the mixture of methanol and H_2O (70:30, V/V).

The detailed methods for characterization and analysis are described in Text S3.

2.5. Metagenomic Analysis. After the experiments, 2 mL of sludge samples were collected from each reactor and subjected to metagenomic analysis by Sangon Biotech (Shanghai) Co., Ltd., China. The detailed procedures are described in Text S4.

2.6. Statistical Analysis. Data were expressed as mean ± standard deviation and derived from three replicate measurements. An ANOVA (Analysis of Variance) test was carried out using R version 4.3.2 to evaluate the significance of the differences. All the statistical differences were deemed significant at $p < 0.05$.

3. RESULTS AND DISCUSSION

3.1. Performances of Methanogenic Digesters. In the first 40 days, glucose was used as the sole substrate, and the influent COD was gradually increased from 2500 to 8000 mg/L. During this stage, the COD removal rate of the IMA group was slightly higher than that of the control group, yet there was no significant difference (Figure 1 A,B and Table S1). From day 41 to day 70, phenol was added to the influent, and its concentration was increased from 500 mg/L (41−58 days) to 1000 mg/L (59−70 days). The average COD concentration in the effluent of the IMA group was 985.70 mg/L, with a removal rate of 87.68%, which was comparable to that of the control group (average effluent COD concentration: 972.92 mg/L, COD removal rate: 87.84%). From day 71 to day124, when the phenol concentration increased to 2000 mg/L, the average effluent COD concentration of the IMA group rose to 1423.62 mg/L, with a removal rate of 82.20%, which was significantly higher than that of the control group (68.57%, average effluent COD concentration: 2514.19 mg/L). Specifically, during the period from 104 to day 124, the COD removal rate of the IMA group was 84.38%, 29.54 percentage points higher than that of the control group (54.84%).

Similarly, when the influent phenol concentration was 500 and 1000 mg/L, both groups demonstrated high phenol removal capabilities. The phenol concentration in the effluent was below 200 mg/L, with a removal rate of over 80%. As the phenol concentration increased to 2000 mg/L, the average phenol concentration in the effluent of the IMA group increased to 531.45 mg/L, which was significantly lower than that of the control group (1008.59 mg/L). Specifically, from day 104 to day 124, the average phenol concentration in the

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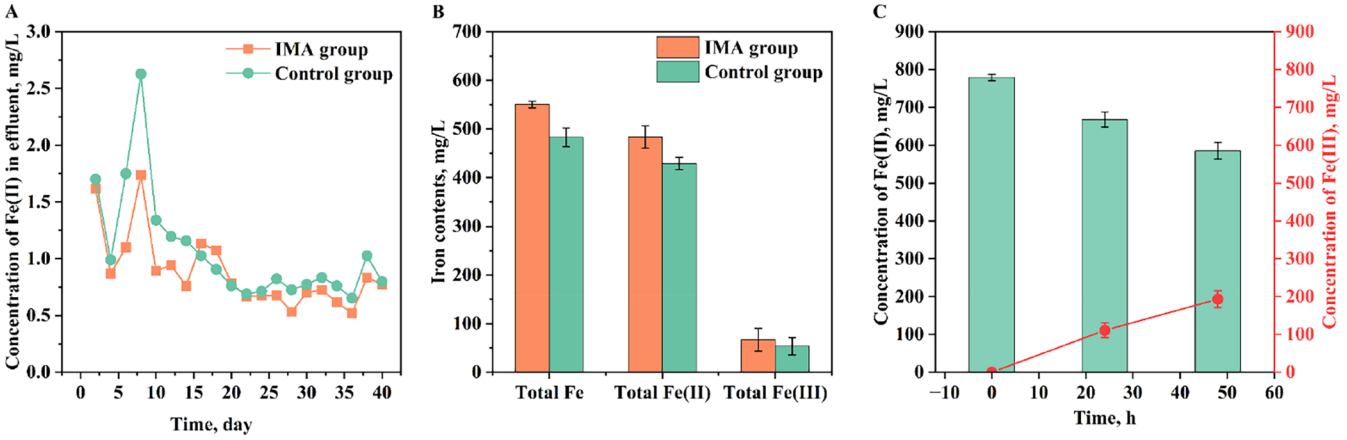


Figure 2. Iron contents. (A) Concentration of Fe(II) in the effluent. (B) Iron contents in the digester after operation. (C) Fe(II) oxidation and Fe(III) generation rate.

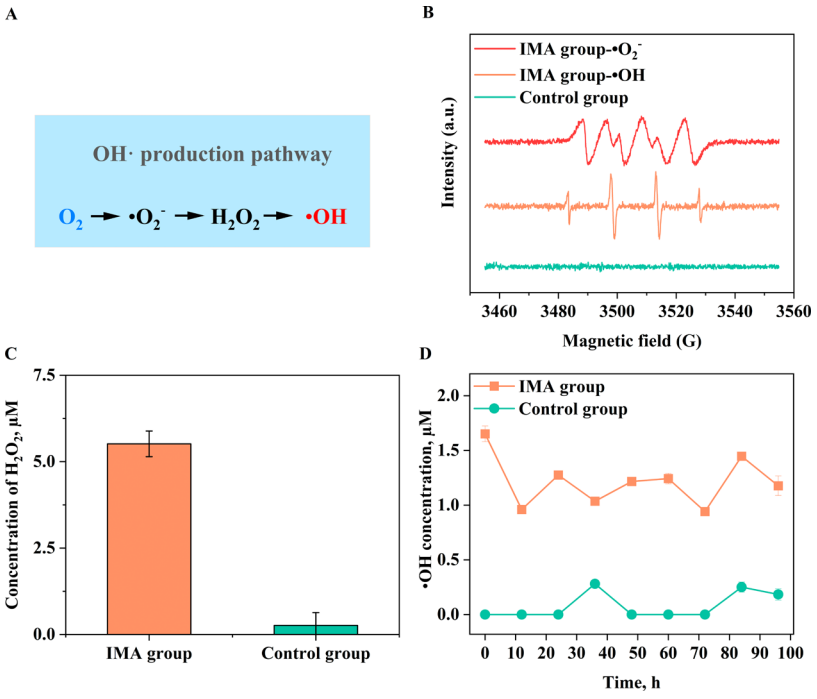


Figure 3. Identification of ROS in intermittent microaeration of anaerobic digestion. (A) Pathway of $^{\bullet}OH$ production. (B) EPR spectra of $^{\bullet}O_2^-$ and $^{\bullet}OH$. (C) H_2O_2 concentration. (D) Time series of $^{\bullet}OH$ generation.

IMA group was 433.73 mg/L, much lower than that in the control group (1427.28 mg/L). Accordingly, the phenol removal rate of the IMA group was 49.68 percentage points higher than that of the control group (78.31 vs 28.64%) (Figure 1 C and Table S1).

The methanogenic capacity of the IMA group was enhanced by intermittent microaeration (Figure 1 D and Table S1). From day 71 to day 124, the average daily methane production of the IMA group was 29.90% higher than that of the control group (1.60 $L/(L_{reactor}\cdot d)$ vs 1.23 $L/(L_{reactor}\cdot d)$). In particular, during the period from day 104 to day 124, the average daily methane production of the IMA group was 1.53 $L/(L_{reactor}\cdot d)$, which was 85.44% higher than that of the control group (0.83 $L/(L_{reactor}\cdot d)$). This might be because intermittent microaeration effectively promoted the degradation of complex substrates, providing suitable substrates for methanogens.[16] Furthermore, in the last 20 days of operation, the methane production per unit COD of the IMA group was 283.34 mL/g COD (1530.2 mL/(8.0 g COD × 84.38% × (24/30)), which was 13.93% higher than that of the control group (248.71 mL/g COD). It indicated that the IMA group had a higher methane conversion efficiency.

3.2. Role of Fe(III)/Fe(II) Cycling and ROS Generation in Phenol Degradation. To verify whether intermittent microaeration drives the Fe(III)/Fe(II) cycle, the iron contents in the effluent and the bulk phases were determined. As shown in Figure 2A, during the first 40 days of operation, the average Fe(II) concentration in the effluent of the IMA group (0.88 ± 0.31 mg/L) was 16.97% lower than that of the control group (1.06 ± 0.47 mg/L). Moreover, the results showed that the total iron concentration (Fe(II) + Fe(III)) in the bulk phase of the IMA group was 550.20 ± 6.99 mg/L, which represented a 13.97% increase compared to the control group (482.78 ± 18.91 mg/L). The total Fe(II) and total

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Fe(III) showed the same trend (Figure 2B). This elevation was likely attributed to the Fe(III)/Fe(II) cycle driven by microaeration.[36,37] Considering that the ferrihydrite was only added at the beginning of the operation, the Fe(II) produced from the DIR might be released to the effluent, reducing the iron content because of the higher solubility of Fe(II) under circumneutral conditions. The lower Fe(II) contents in the effluent and the higher iron content in the sludge of the IMA group indicated that the oxidation of Fe(II) occurred with intermittent microaeration.

To verify this, 1 L of the supernatant was collected from the effluent, and 15 mM $Fe(OH)_2$ was added. When the depletion of organic matter failed to provide adequate electrons for DIR, intermittent microaeration was initiated under the same conditions as those in the above continuous flow experiment, and the oxidation rate of Fe(II) was determined. The results showed that intermittent microaeration could effectively oxidize Fe(II), and an average of 1.86 mM of Fe(II) was oxidized every 24 h, accompanied by the formation of Fe(III) (Figure 2C). The formation of Fe(III) reduced the loss of iron because its solubility was lower than that of Fe(II). Furthermore, the oxidation of Fe(II) led to a continuous supply of Fe(III), which drove DIR to degrade organic matter and enrich iron-reducing bacteria. Iron-reducing bacteria could obtain energy by reducing Fe(III), thus enhancing the degradation of organic matter.[38] Lovley et al. have demonstrated that glucose fermentation combined with Fe(III) reduction (−76 kJ/mol) was energetically more favorable than without the reduction of Fe(III) (−71 kJ/mol).[39]

In addition to the function of DIR, the oxidation of Fe(II) may induce the Haber-Weiss mechanism (reactions 2−5) to generate ROS such as $^{\bullet}O_2^-$, H_2O_2, and $^{\bullet}OH$ for the degradation of substrate (Figure 3A). During the oxidation of Fe(II), $^{\bullet}O_2^-$ and H_2O_2 are important intermediate species, and H_2O_2 is also a precursor for the production of $^{\bullet}OH$ through the Fenton-like reaction.[32] The $^{\bullet}O_2^-$ was detected by EPR with 5,5-dimethyl-1-pyrroline-*N*-oxide (DMPO) as a trapping agent. As shown in Figure 3B, the IMA group exhibited a typical DMPO/$^{\bullet}O_2^-$ hexagonal peak (with a peak intensity ratio of 1:1:1:1, α_N = 13.36 G, α_H^{β} = 14.06 G),[40] while the control group did not show a similar peak. Furthermore, the amount of H_2O_2 generated after the microaeration was determined spectrophotometrically using cerium sulfate. The results showed that the average concentration of H_2O_2 in the IMA group was 15.54 ± 2.31 μM, whereas it was almost undetectable in the control group (Figure 3C).

DMPO was used as a trapping agent to identify $^{\bullet}OH$. As depicted in Figure 3B, a characteristic signal of the DMPO/$^{\bullet}OH$ complex (with a peak intensity ratio of 1:2:2:1, α_N = 14.89 G, α_H^{β} = 14.89 G) was detected in the mud-water mixture of the IMA group, while this characteristic peak was absent in the control group. To determine the concentration of $^{\bullet}OH$, 50 μM coumarin was added to the influent to capture the $^{\bullet}OH$ in situ, leading to the generation of 7-hydroxycoumarin.[32,41] The results showed that there was a stable generation of $^{\bullet}OH$ in the IMA group (1.22 ± 0.04 μM), whereas almost no $^{\bullet}OH$ was detected in the control group during the consecutive four-day monitoring period. To verify the generation of $^{\bullet}OH$, alcohols (TBA and NBA) were used as $^{\bullet}OH$ quenchers, and both of them exhibited significant quenching effects on $^{\bullet}OH$ formation (Figure S2).

Furthermore, the generation pathway of $^{\bullet}OH$ was determined through quenching experiments using NBT and CAT. NBT acts as an efficient scavenger of $^{\bullet}O_2^-$, while CAT can decompose H_2O_2, thereby suppressing $^{\bullet}OH$ generation.[32,42] Both NBT and CAT exhibited a significantly inhibitory effect on the production of $^{\bullet}OH$ (Figure S2). These results suggested that $^{\bullet}O_2^-$ and H_2O_2 served as intermediates in the production of $^{\bullet}OH$, and the production pathway of $^{\bullet}OH$ was through the successive reduction of molecular O_2 ($O_2 \rightarrow {}^{\bullet}O_2^- \rightarrow H_2O_2 \rightarrow {}^{\bullet}OH$) (Figure 3A).[23] H_2O_2 and $^{\bullet}OH$ were detected during anaerobic digestion, indicating that intermittent microaeration could induce the generation of ROS through a Fenton-like reaction.[23,43] This might be an important reason for the rapid degradation of phenol in the IMA group.[44]

The degradation of phenol facilitated by DIR, in which iron-reducing bacteria use Fe(III) oxides as the final electron acceptors for phenol oxidation, has been proven to be an efficient method for enhancing its anaerobic biodegradation.[3,45] Besides, DIR can facilitate hydrolytic acidification and utilize simple substrates like glucose.[31] Since both the IMA group and the control group were initially dosed with ferrihydrite (15 mM Fe(III)), the degradation of glucose and phenol in the two reactors was initially effective, and the methane production was similar. However, when the phenol concentration increased to 2000 mg/L, the control group was unable to efficiently metabolize phenol and tended to collapse, while the IMA group was still able to maintain a phenol removal rate of 78.31% and stable methane production. This might be attributed to the introduction of microaeration, which drove the Fe(III)/Fe(II) cycle and generated ROS to accelerate the degradation of phenol.

To further verify the effects of DIR and ROS generation on phenol degradation, another experiment was carried out with four groups using phenol (350 mg/L) as the sole carbon source: Group A without ferrihydrite addition and microaeration, Group B with ferrihydrite addition, Group C with ferrihydrite addition and microaeration, and Group D with ferrihydrite addition and microaeration, and formate (5 mM) as $^{\bullet}OH$ quencher (D group). The amounts of ferrihydrite added (15 mM Fe(III)) and the microaeration pattern (14 mL $O_2/(L_{reactor}\cdot day)$) were the same as the UASB digester.

The results showed that both ferrihydrite addition and microaeration contributed to phenol degradation (Figure S3). For example, on the sixth day of the reaction, the highest removal rate of phenol was 65.41% in Group C, which was higher than that in Group B (42.63%) and Group A (38.15%). The difference between Group A and Group B might be due to the effect of DIR, where phenol degradation was accompanied by a large amount of Fe(II) production (Figure S3B).[3] The contribution of DIR to promoting phenol degradation accounted for approximately 23.41% ((42.63−38.15%)/(65.41−38.15%)). After adding formate as the $^{\bullet}OH$ quencher, the phenol degradation rate in Group D was 61.51%, lower than that in Group C (65.41%). It indicated that $^{\bullet}OH$ played a role in phenol degradation, accounting for about 14.31% ((65.41−61.51%)/(65.41−38.15%)) of the increased phenol degradation.

The degradation of phenol proceeds through different processes depending on aerobic or anaerobic conditions. Under aerobic conditions, phenol is decomposed into acetic acid and succinic acid, which are subsequently mineralized into CO_2 and H_2O via the tricarboxylic acid cycle. In anaerobic

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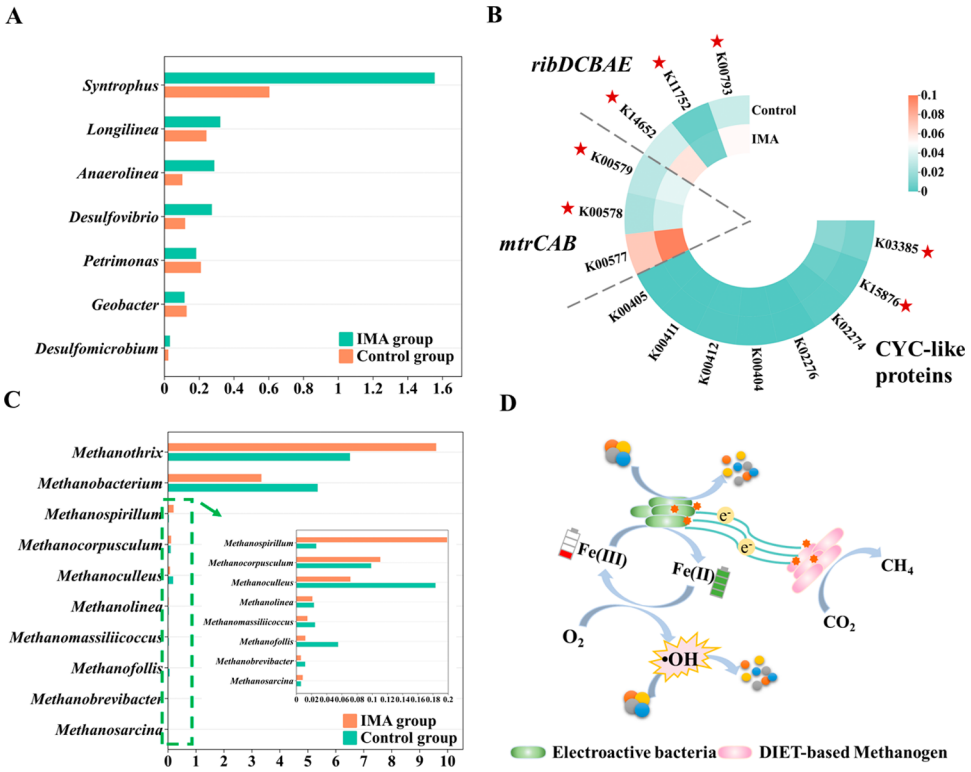


Figure 4. (A) Relative abundance of bacterial community at the genus level. (B) Abundance of genes related to dissimilatory iron reduction. (C) Relative abundance of archaeal community at the genus level. (D) Schematic model of enhancing anaerobic digestion through the Fe(III)/Fe(II) cycle and the generation of reactive oxygen species under intermittent microaeration conditions. The red stars indicated significant differences between the two groups ($p < 0.05$).

environments, phenol is transformed into benzoic acid and further metabolized to acetic acid and is ultimately mineralized into CH_4 and CO_2.[46,47] Compared with anaerobic conditions, the degradation rate of phenol is faster under aerobic conditions,[46] and introducing micro-oxygen into anaerobic digestion is beneficial for phenol degradation. However, aerobic respiration resulting from microaeration is not the main reason for the enhanced phenol degradation.[48] In this study, even if all the supplied O_2 (14 mL/($L_{reactor}$·day)) was used for COD removal, it could only increase the COD removal by 40 mg/L (14 mL × 0.001429 g/mL (SATP) × 2), which only accounted for 1.7% of the observed increase in phenol removal (2365 mg/L COD = (1566.27−572.72 mg/L) × 2.34). This is because the ROS generated from the reduction of O_2 likely participated in the ring-cleavage degradation of phenol molecules, producing intermediates that can be utilized for methanogenesis.

3.3. Microbial Community Composition of Sludge. The sludge collected from the reactors after 124 days of operation was analyzed through Metagenomic analysis. There was a distinct separation of the microbial communities between the IMA group and the control group (Figure S4). Intermittent microaeration enriched dissimilatory iron-reducing bacteria (Figure 4A). The abundance of dissimilatory iron-reducing bacteria (*Anaerolinea*, *Desulfomicrobium*, *Desulfovibrio*, *Geobacter*, *Longilinea*, *Petrimonas*, and *Syntrophus*) in the IMA group (2.77%) was 1.94 times higher than that in the control group (1.43%) ($p < 0.01$).[49,50] *Syntrophus* was the predominant dissimilatory iron-reducing bacteria, which was significantly enriched in the IMA group with a relative abundance of 1.56%, much higher than the control group (0.60%) ($p < 0.01$). Besides, compared with the control group, *Anaerolinea* (0.29 vs 0.10%, $p < 0.01$) and *Desulfovibrio* (0.27 vs 0.12%, $p < 0.01$) were also significantly enriched in the IMA group.

Desulfovibrio, a sulfate-reducing bacterium, can transfer extracellular electrons to electrodes to generate electricity and is one of the crucial electroactive bacteria involved in DIET.[49] Isolated *Desulfovibrio* strains could grow through DIET when cocultured with *Methanothrix harundinacea* and *Methanobacterium* strain YSL.[51] On the other hand, *Desulfovibrio* can highly express the genes encoding cytochrome c (CYC)-like proteins. The results of metagenomic analysis showed that the genes encoding cytochrome c (CYC)-like proteins in the IMA group were 1.50 times that in the control group (Table S2) ($p < 0.01$).[49,52]

Furthermore, the expressions of other genes related to DIR, including metal-reducing conduit biosynthesis gene cluster *mtrCAB* and flavin biosynthesis gene cluster *ribDCBAE* were analyzed.[53] MtrA, MtrB, and MtrC form a trans-outer membrane protein complex that mediates electron transfer from periplasmic proteins to the bacterial surface, enabling extracellular electron transfer (EET) through direct interaction with external electron acceptors.[54] Besides, redox-active electron shuttles, such as flavins, are well-established mediators of EET during microbial respiration, and an increase in their concentration can significantly enhance the EET rates.[53,55] The results showed that the genes related to *mtrCAB* cluster (0.29 vs 0.21) and *ribDCBAE* cluster (0.12 vs 0.07) in the IMA group were 1.42 and 1.67 times those in the control group, respectively (Figure 4B and Table S2). The increased expression of genes related to DIR indicated an enhanced extracellular respiration capacity in the IMA group. Therefore,

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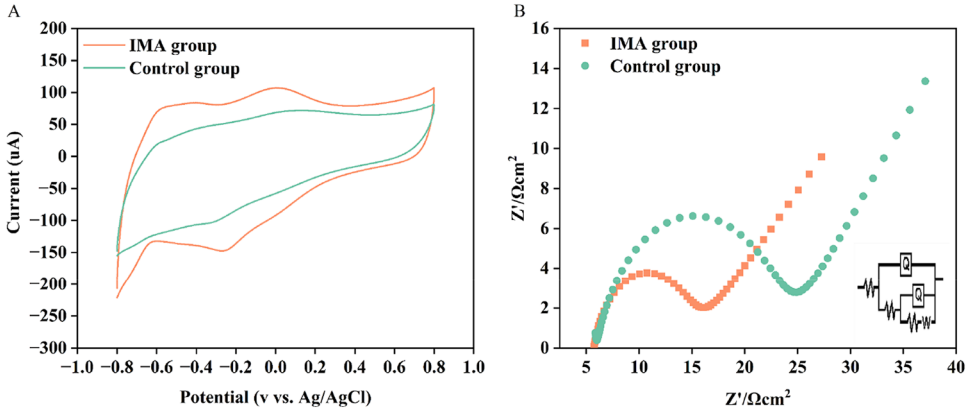


Figure 5. Changes in microbial electrical activity in the sediment. (A) Curve of cyclic voltammetry. (B) Curve of electrochemical impedance spectroscopy.

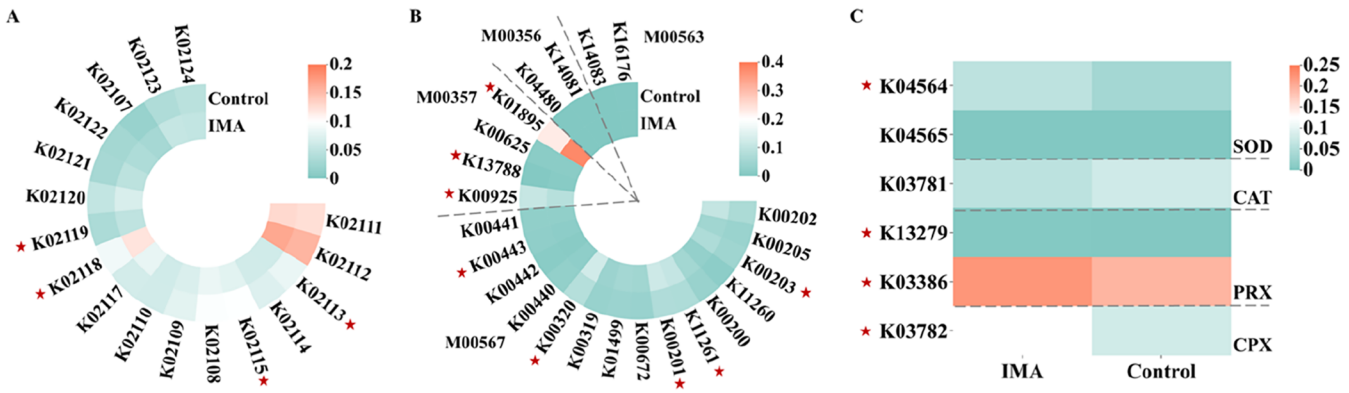


Figure 6. (A) Abundance of genes that encode ATPase. (B) Abundance of genes that encoded the key enzymes involved in four methanogenic pathways in the methanogenic sludge. (C) Abundance of genes that encode the ROS detoxification in the methanogenic sludge. The red stars indicated significant differences between the two groups ($p < 0.05$).

the electrons derived from the oxidation of organic matter could be more efficiently transferred to extracellular acceptors, thereby promoting the generation of extracellular ROS.

Intermittent microaeration enriched methanogens capable of DIET. *Methanothrix* is a typical acetate-utilizing methanogen and a typical methanogen capable of accepting electrons via DIET to reduce CO_2 to CH_4.[49,56] Compared with the control group, *Methanothrix* was significantly enriched in the IMA group (9.59 vs 6.51%) (Figure 4C), suggesting that microaeration is favorable for their enrichment, which also indicated that it is tolerant to O_2. In addition, *Methanospirillum*, a typical hydrogen-utilizing methanogen with the potential to form DIET with the electroactive microorganism *Syntrophus*, also showed a 6.7-fold increase in relative abundance in the IMA group (0.20 vs 0.03%).[57]

The enrichment of iron-reducing bacteria and methanogens capable of DIET indicated that the IMA group might have established an efficient methanogenic pathway via DIET to promote methane production, which was consistent with the higher methane production rate in the IMA group. Intermittent microaeration could oxidize Fe(II) to Fe(III) in the anaerobic digestion system, thereby providing a continuous supply of Fe(III) for DIR and facilitating the enrichment of iron-reducing microorganisms, which furthermore could construct a DIET with methanogens to promote methanogenesis (Figure 4D).

3.4. Enhancement of Electrochemical Activity. The electroactivity of the microbial community was significantly enhanced under intermittent microaeration conditions. The curve of cyclic voltammetry (Figure 5A) showed that the capacitance of the sludge in the IMA group was 1.60×10^{-3} F, which was 48.14% higher than that of the sludge in the control group (1.08×10^{-3} F). This indicated that more electron storage sites were involved in the electron exchange between microorganisms and extracellular electron acceptors.[3,31]

In addition, the results of electrochemical impedance spectroscopy showed that the resistance of sludge in the IMA group was 11.11 Ω, 40.39% lower than that of the sludge in the control group (18.64 Ω) (Figure 5B). The decrease in resistance suggested an increased rate of electron transfer within the sludge. Resistance and capacitance are frequently used as indicators to evaluate the extracellular respiration activities of microbial communities.[31,34] The lower resistance and higher capacitance suggested that intermittent microaeration improved the electroactivity of the microbial community and facilitated extracellular electron transfer. This might be attributed to the Fe(III)/Fe(II) cycle, which was beneficial for the enrichment of electroactive microorganisms.

3.5. Gene Expression for Microbial Activity and Metabolism. The enzymatic activities of microorganisms were characterized to understand the microbial factors associated with the enhanced performance of anaerobic digestion by the Fe(III)/Fe(II) cycle with intermittent microaeration. Fe(III) served as an electron acceptor, facilitating extracellular respiration and promoting ATP production in microorganisms. According to the metagenomic

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analysis results, the abundance of genes encoding ATPase (EC: 7.1.2.2, 7.1.2.1) in the IMA group was 1.24, significantly higher than that in the control group, which was 1.16 ($p < 0.05$) (Figure 6A and Table S3)

During anaerobic digestion, electron acceptors (such as the reduction of H^+ to H_2) are necessary for acidogenic bacteria to break down complex organic substrates. Additionally, methanogens rely on the continuous utilization of the small molecule acids and H_2 generated by acidogenic bacteria to keep the pH within a suitable range, which is essential for the continuous decomposition of organic substances.[58] Methanogens act as the ultimate electron acceptors in the degradation of organic matter during the metabolic exchange between acid-producing bacteria and methanogens.[59] The Fe(III) generated by intermittent microaeration could replace methanogens as an extracellular electron acceptor for acidogenic bacteria, constantly facilitating extracellular respiration and ATP synthesis.[38] This could explain the improved removal of COD facilitated by the Fe(III)/Fe(II) cycle.

To evaluate the methanogenic capacity of methanogens, the abundance of genes encoding key enzymes involved in four methanogenic pathways, namely carbon dioxide reduction [M00567], acetate decarboxylation [M00357], methanol methanogenesis [M00356], and methylamine/dimethylamine/trimethylamine-type methanogenesis [M00563], was analyzed based on the metagenome (Figure 6B and Table S4).[56] Among these, the reduction of carbon dioxide and the decarboxylation of acetate were the dominant methanogenic pathways, while the abundance of genes encoding key enzymes for the other two methanogenic pathways was relatively low.

The genes responsible for encoding the key enzymes involved in carbon dioxide reduction were significantly more abundant in the IMA group (0.87) than in the control group (0.63) ($p < 0.05$). Additionally, the genes encoding the key enzymes for acetate decarboxylation in the IMA group (0.49) were notably increased, being 1.36 times as prevalent as those in the control group (0.36). In addition, the abundance of genes encoding the first two enzymes of the carbon dioxide reduction pathway, including those encoding formyl-methanofuran dehydrogenase (fwdA, fwdB, fwdC, fwdD, and fwdE) and 4Fe-4S ferric ferredoxin (fwdF and fwdG) (0.53), was significantly higher than that of the gene encoding first enzyme in the acetate decarboxylation pathway (acetyl-CoA decarboxylase/coenzyme, 0.38) in the IMA group ($p < 0.05$). This suggested that the related methanogens (i.e., *Methanothrix*) might be actively engaged in reducing CO_2 to CH_4, i.e., *Methanothrix* produced methane through DIET.[56]

During the process of biological consumption, molecular O_2 accepts electrons to generate $^{\bullet}O_2^-$, along with H_2O_2 and $^{\bullet}OH$, which can be harmful to microorganisms. Aerobic bacteria generally possess antioxidant defense mechanisms that counteract and alleviate the detrimental effects caused by an overload of ROS by producing a range of antioxidant enzymes.[13] It is well-known that obligate anaerobes lack such a defense system and are unable to survive under oxidative stress. Recent studies have shown that some methanogenic genera, such as *Methanosarcina* and *Methanocella* can cope with oxidative stress by encoding a variety of antioxidant enzymes (such as superoxide dismutase, superoxide reductase, catalases, peroxidases, peroxiredoxins, and rubrerythrin) for detoxification purposes.[13] The results of microbial community analysis showed that the abundances of *Methanosarcina* and *Methanocella* in the IMA group were 1.38 and 1.23 times those in the control group, respectively.

Furthermore, it has been reported that *Methanothrix* can gradually establish O_2 tolerance in a microaerobic environment at an appropriate aeration intensity.[60,61] However, there is limited information available regarding its O_2 tolerance mechanisms and changes in the activities of related enzymes. In this study, *Methanothrix* was enriched and became dominant in the IMA group under an aeration intensity of 14 mL O_2/($L_{reactor}$·day), indicating that this aeration level was within its tolerable range. In addition, the metagenome results showed that the levels of superoxide dismutase (EC 1.15.1.1: 0.06 vs 0.03), catalases (EC 1.11.1.6: 0.14 vs 0.12), peroxiredoxins (EC 1.11.1.15: 0.23 vs 0.20), and catalase-peroxidase (EC 1.11.1.21: 0.13 vs 0.08) were higher in IMA groups than in the control group (Figure 6C and Table S5). Among them, genes encoding superoxide dismutase and peroxiredoxins are mainly originated from *Methanothrix*. This indicated that *Methanothrix* could effectively scavenge intracellular $^{\bullet}O_2^-$ and H_2O_2, protecting cells from oxidative stress damage.

The enrichment of methanogenic genera capable of coping with oxidative stress and the elevated abundance of antioxidant enzyme genes might explain why methanogenesis could proceed effectively under intermittent microaeration conditions. This confirmed that the ability of anaerobic sludge to cope with oxidative stress could be domesticated by O_2 induction within the anaerobic digestion system, which is consistent with the previous studies.[62,63]

Furthermore, in iron-containing anaerobic digestion systems, intermittent microaeration might oxidize Fe(II) to drive the Fe(III)/Fe(II) cycle, which could enrich DIR bacterial and enhance EET capacity. This mechanism preferentially directs O_2 reduction in the extracellular space to form extracellular ROS, accelerating the degradation of the substrate and thus minimizing the generation of intracellular ROS and its associated detrimental effects. Reducing the permeability of O_2 into cells can alleviate the oxidative stress induced by it and enhance O_2 tolerance of anaerobic microorganisms.[64]

4. IMPLICATIONS

It has been reported that phenol with a concentration exceeding 2000 mg/L can inhibit biological processes, including anaerobic digestion, due to the toxic accumulation of phenol.[6,7] Therefore, pretreatment is a necessary step before the biological treatment of phenolic wastewater. This study demonstrated that the Fe(III)/Fe(II) cycle and ROS generation induced by intermittent microaeration significantly enhanced phenol degradation. These findings suggested the potential to reduce or even eliminate conventional pretreatment processes, thus reducing the operational costs of anaerobic treatment systems.

A recent study has reported the generation of ROS in activated sludge systems,[23] however, the influence of ROS generation on anaerobic digestion and methanogenesis remains unclear. This study investigated the ROS generation induced by microaeration in anaerobic digestion and its promotion of anaerobic performance. The intracellular ROS scavenging mechanism enabled methanogens to survive under intermittent microaeration. In addition, changes in the redox potential of the environment can affect the distribution of intracellular and extracellular ROS, as well as the redox state of the cell, thereby regulating the associated metabolic processes.[62] Therefore, the adaptation of methanogens to

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varying aeration levels and the distribution of intracellular/extracellular ROS warrant further investigation.

Furthermore, it should be noted that iron is a commonly used additive in pretreatment or advanced treatment processes within wastewater treatment systems.[65,66] In these treatment systems, microaeration of the anaerobic methanogenic or hydrolytic acidification units was expected to improve treatment efficiency and enhance the degradation of biologically refractory substances.[13,15] However, the reasons were generally attributed to the enhancement of microbial community diversity, and there is a lack of explanations from the perspective of intrinsic mechanisms. This study provided a new perspective for improving anaerobic systems with intermittent microaeration and an effective method with economic feasibility for upgrading existing wastewater treatment systems (Text S5). This might also explain the enhanced decomposition of pollutants in artificial and natural biofilm systems in terms of chemical mechanisms, i.e., O_2 gradient results in an anaerobic environment inside the biofilm and aerobic environment outside the biofilm, which could potentially accelerate pollutant decomposition through the generation of ROS via Fenton-like reactions.[23] This hypothesis requires further investigation.

■ ASSOCIATED CONTENT

Supporting Information

The supporting information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.est.5c04187.

Experimental methods; performance parameters of the reactor during operation; the typical genes involved in dissimilatory iron reduction and their abundance based on KEEG; the $^{\bullet}OH$ production in the presence of NBT (Nitro blue tetrazolium), CAT 174(Catalase), TBA (*tert*-butanol), and NBA (*n*-butanol) (PDF)

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Author Contributions

X.W.: Investigation, formal analysis, and writing—original draft. J.Z. and H.M.: Investigation. B.Y.: Methodology. Q.Y.: Writing—review and editing. Y.Z.: Conceptualization, writing—review and editing, and funding acquisition.

Notes

The authors declare no competing financial interest.

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